

Normalization → also known as MinMax Scaling

↓
It shifts and rescales numerical data so that all values fall within a strict, bounded range — most commonly 0 to 1 (sometimes -1 to 1)

↓
Unlike Standardization, which leaves data

boundaries open-ended, normalization forces every single point into a fixed box

New normalized value

Original raw value

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Types

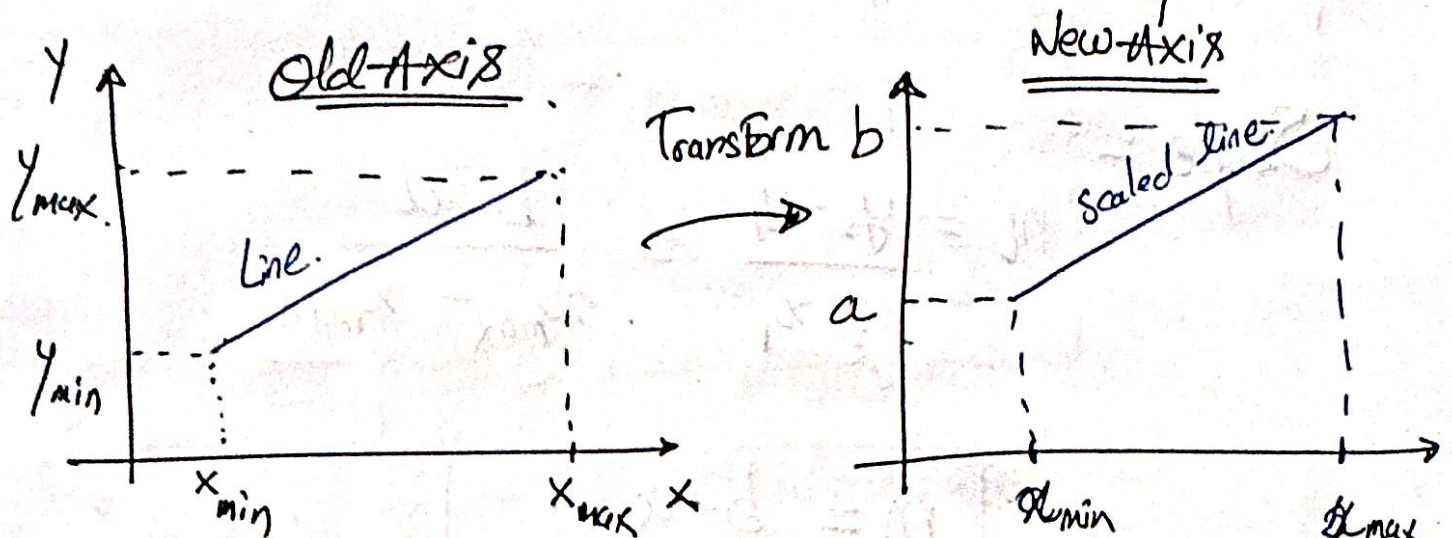
↓
Standard Min-Max scaling (0 to 1)

↓
Custom Range Min-Max Scaling (eg -1 to 1)

↓
Max Absolute Scaling (MaxAbsScaler)

a) Standard Min-Max Scaling (0 to 1)

In geometry, if a line segment of certain length, and want to shrink, stretch or slid it onto a new track without bending it for changing the relative spacing b/w points along the line $\rightarrow$ use affine map.



NOTE - during transformation the relative position x in its original range must be exactly equal its relative position in the new range.

using Coordinate Geometry :-

$$y = mx + c$$

$\downarrow$ $\downarrow$
Scaled output raw input

Coordinate Check points

1) when input is minimum,
Output must be $a : (x_{\min}, a)$

2) when input is maximum,
Output must be $b : (x_{\max}, b)$

Slope :

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{b - a}{x_{\max} - x_{\min}}$$

$$m = \frac{b - a}{x_{\max} - x_{\min}}$$

y-intercept :

$$y = mx + c$$

$$a = \left(\frac{b - a}{x_{\max} - x_{\min}} \right) x_{\min} + c$$

$$c = a - \left(\frac{b - a}{x_{\max} - x_{\min}} \right) x_{\min}$$

So, $y = \underline{mx} + \underline{c}$

$$y = \left(\frac{b-a}{x_{\max} - x_{\min}} \right) x + a - \left(\frac{b-a}{x_{\max} - x_{\min}} \right) x_{\min}$$

$$y = (b-a) \left(\frac{x - x_{\min}}{x_{\max} - x_{\min}} \right) + a$$

$$\frac{y-a}{b-a} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

If target range is standard (0 to 1)
So, $a=0$ and $b=1$

$$\frac{y-0}{1-0} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

$$y = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

(CM)

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

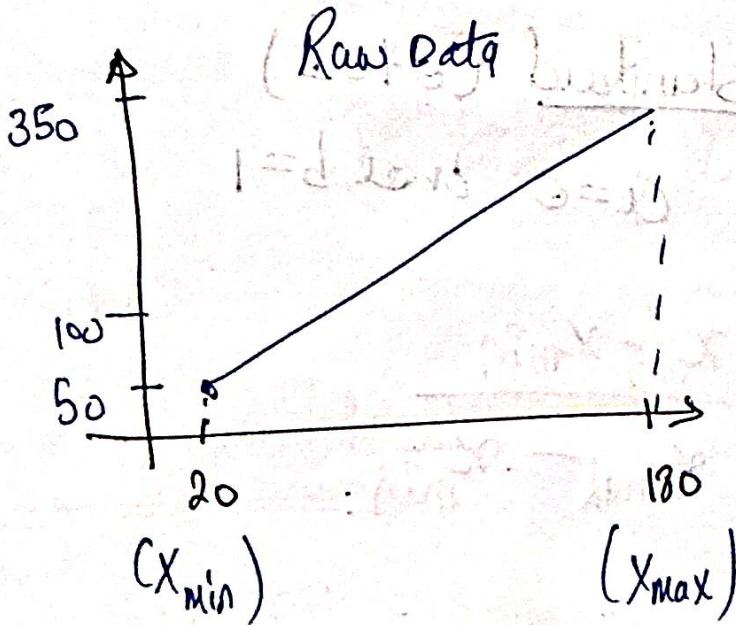
$x = x_{\min} \Rightarrow x_{\text{norm}} = 0$
 $x = x_{\max} \Rightarrow x_{\text{norm}} = 1$
 so, $[0, 1]$

e.g. If the ground truth can be found with

$$y = 20x + 2$$

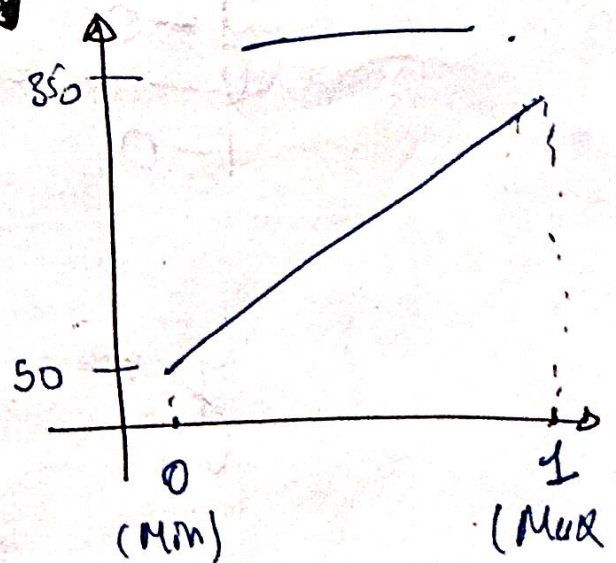
$x \rightarrow$ raw data

↓
After Min-Max Scaling $\rightarrow x_{\text{norm}}$



$x \rightarrow x_{\text{norm}}$

Calculate New
 $y = 20x_{\text{norm}} + 2$
 Bounded Axis



When to use:

Highly effective when

data

Math

Operation
requires
specific input
domain.

boundaries are
strictly defined

1) Image Processing & Computer Vision:

Image Pixel: 0 to 255
(dark) (white)

→ Normalizing them to $[0, 1]$ or $[-1, 1]$
range standard for CNNs

2) Bounded Activation Functions:

ANN that use bounding activation
function like

Sigmoid
(output $[0, 1]$)

Tanh
(output $[-1, 1]$)

Train Much faster when input
layers share those
exact same ranges.

3) Distance-Based Algorithms with No Distribution Assumptions:

K-NN, K-Mean, Clustering

If Not Know the underlying distribution of data
↓

Min-Max Scaling puts all features on an even playing field without assuming bell curve.

4) Interpretable Direct Comparison:

For Scoring engine (e.g credit score)

bounded no. b/w 0 to 1 → Easier for Humans to read

↓
Convert into %

then z-score.

when to avoid :-

1) when Outliers are present:

If feature contains extreme values, Min-Max Scaling

Compress the remaining "normal" data into tiny fraction cluster.

→ destroying model's ability to differentiate b/w them.

2) when using Variance

driven Algo; like PCA → rely heavily on finding the direction of max variance.

→ Min-Max Scaling alters the variance of features uniformly.

3) when Data Appears Frequently Outside the bounded Range;

which can distort the true principal component.

If test data or in production → introduces value smaller than x_{min} or larger than x_{max} .

Normalized values will overflow outside $[0, 1]$ boundaries → causing instability

Architectural Remarks:-

1) Preservation of Zeros :- If data set contains true zeros (absence of something)

Min-Max scaling will keep those zeros.

"Only if the minimum value of that feature is 0"

"If x_{min} is a negative value true zeros will shift"

2) No Shape Distortion: If data is heavily skewed
↓
remains heavily skewed after scaling
"If only changes the Axis"

3) Pipeline Longevity:-
Step 1 Calculate $[X_{min}]$ and $[X_{max}]$
from training set.
↓
Use same for input data in production

b) Custom Range Min-Max Scaling

As we have

$$\frac{y-a}{b-a} = \frac{x-x_{\min}}{x_{\max}-x_{\min}}$$

$$y = a + \frac{(x-x_{\min}) \cdot (b-a)}{(x_{\max}-x_{\min})}$$

Slide to target minimum
0 to 1

Change length of data

$$x_{\text{norm}} = a + \frac{(x-x_{\min}) \cdot (b-a)}{(x_{\max}-x_{\min})}$$

Standard normalization
(convert into 0 to 1)
Strength from 0 to (b-a)
a b

if range is -1 to 1 so, $(b-a) = 1 - (-1) = \underline{\underline{2}}$

When to use:-

1) Hyperbolic Tangent (Tanh):

Deep neural network $\xrightarrow{\text{use}}$ Tanh

↓
Improve Performance
if input is in
same range.

2) Audio Signal Processing: Sound waves data and digital audio arrays are natively centered around zero.

→ Scaling audio amplitude features strictly to $[-1, 1]$

3) Image Processing (Pre-Trained Model): Preserves sound wave phase.

Advance Computer vision Model
(MobileNet v2...)
use image pixel range $[-1, 1]$.

When to avoid?

1) when Dataset has Extreme Outliers
as use x_{min} and x_{max} even one outlier
can skew 99% of data removing clean data will get crushed into a tiny Area.

2) when preserving "True Zero" is critical
(Sparse Data)

If we scale data set containing structural zeros
(like text word-count, one-hot encoding, Sparse-user ratings)

(value into range like $[-1, 1]$ $\rightarrow$ Our zeros will Shift.)

- Unless Original $x_{\min}$ was exactly 0 and our range starts at 0, a raw value of 0 $\xrightarrow{\text{Transf}}$ non-zero No. (eg -0.5)

- This destroys the "Sparsity" of data forcing Computer to calculate some which was Memory-efficient zero.

3) When Production Data is unpredictable
(Out of bound/risk)

$\rightarrow$ A new max value will overflow boundary.
resulting scaled value like 1.4 or 2.1
instead of $\rightarrow \underline{1.0}$ value

$\rightarrow$ If neural network relies on strict $[-1, 1]$
Constraints. to prevent mathematical overflow.

these boundary-breaking values can cause
training or prediction instability.

4) Using Variance
driven by: PCA.

c) Max Absolute Scaling (Max Abs Scale)

↳ specialized normalization technique that scales features by dividing every value by its ^{max.} absolute value.

"This Forces all the data to Fall Strictly within a bounded range of -1 to 1, while keeping the data perfectly centred at its origin position."

$$X_{\text{scaled}} = \frac{x}{\underbrace{\max(|x|)}_{\text{Max absolute of Particular feature}}}$$

When to use :-

1) Sparse Text Data (NLP) :-

when converting text documents into numbers
using techniques like TF-IDF or
Bag-of-words.
get giant table containing
Millions of rows $\rightarrow$ 99% zeros
MaxAbsScaler $\rightarrow$ scales the word
frequency down safely
without touching the
zeros.

2) Centered
Signal Processing :-

Data like electrical voltage, microphone
audio, sensor vibration

$\downarrow$
they naturally
fluctuate around
zero.

MaxAbsScaler

$\rightarrow$ preserve the zero
and clipping peaks at 1.0
and
troughs at -1.0

3) Linear Bounded
Models :-

works good where algorithm
wants input as $\in [-1, 1]$.

(like Neural Network with tanh)

$\downarrow$
when data is
Sparse.

When to avoid:-

1) Extreme Outliers:

As it relies on a single global maximum ($\max(1000)$), a massive outlier will ruin the scale.

Noise $\rightarrow 10000$

While other data points around $\rightarrow 10$

↓
will squash the data.

2) Fully Dense, Strongly Skewed Data:

If data does not have zeros and does not naturally center around zero.

↓
Standard Min-Max Scaling or Standardization will yield better performance.

Robust Scaling

different from Standard Normalization and Standardization

"Rather than relying on global limits (normalization) or avg (standardization),

Robust Scaling utilizes order-based handle datasets Percentiles to
! plagued by extreme noise and heavy outliers.

Normalization (Min-Max): use (x_{min}, x_{max}) limits

A single extreme outlier entirely breaks the scale.

Standardization (z-score): use Mean(μ) & SD(σ)

→ As Mean(μ) & σ use every data points

→ extreme outliers heavily distort these parameters

Robust Scaling:

use Median &

Interquartile Range (IQR)

If completely ignores the values of the outer edges, focusing on stable centre.

For every individual value (x) in Feature
Robustly Scaled Value (X_{robust}) is given as:

$$X_{\text{robust}} = \frac{x - \text{Median}}{\text{IQR}}$$

Where

$$\text{IQR} = Q_3 - Q_1$$

- Median (Q_2): The exact Middle value (50th percentile) when data is ordered.
- Q_1 (25th Percentile): The value below which 25% of data falls
- Q_3 (75th Percentile): The value below which 75% of data falls.

The Deciding Framework

Normal
(Gaussian) Distribution

- Use Standardization $\rightarrow$ If data looks like a bell curve or algorithm relies on data variance.
(Preserve outliers)
- Use Min-Max Scaling $\rightarrow$ If need strict, bounded limits (Image Pixels) or using Neural Network
(Normalization)
- Use Robust Scaling $\rightarrow$ If data is noisy and contains extreme, unpredictable outliers.

Is data plagued by extreme outliers?

Yes No

Robust Scaling

Does algorithm requires a strict bounded range?

Yes No

MinMax Scaling

Standard Scaling

① Standardization (Z-Score)

Choose this as your absolute default
For most classical ML Algo

Data Profile: When features roughly follow
Normal (Gaussian) Distribution
with few or no extreme
Outliers

Best For

- Linear Models where Coefficients
represents feature importance

Linear
Regression

Logistic Regression.

- Geometric, Variance-driven algo.

PCA

SVM

Why?

1) It Preserves outliers without
Crushing the normal data

2) Centering features at 0 make
gradient descent
Optimize much faster

② Min-Max Scaling (Normalization)

Choose when algorithm explicitly demands or performs better with strict bounded bounding box

Data Profile: The absolute max and min limits of data and distribution is completely non-gaussian

Best Used for

a) Image Processing: natively pixels 0 to 255
↓

b) Deep Learning: map to $[0, 1]$
Output/Input layers: Neural networks

Using bounded activation function like
Sigmoid $[0, 1]$
or
Tanh $[-1, 1]$

c) Distance-based algo like KNN or K-means Clustering

when data is clean,
Outlier-free

⑤ Robust Scaling

Choose when data contains lots of Outliers that can't easily clean up.

Data Profile: when data $\rightarrow$ heavy noise or massive outliers

Best used for:

a) Financial and Fraud Detection

Datasets: Transaction amounts naturally have massive, valid spikes

b) IoT and Industrial Sensor Logs

Why? If scales the "normal" middle 50% of data while leaving the Outliers out in wings.

↓
Keeping them machine-readable without skewing the scaling of regular data.